t2-w09 — Simplification of Algebraic Terms — Simplification of algebraic expressions: order of operations, brackets and word problems

Run `diag-v3` · conditions C0 and C1 · the pack shown is draw 3 of each.

C0 `The curriculum objective alone. No lesson plan, no note.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Simplification of Algebraic Terms

`subtopic`

Simplification of algebraic expressions: order of operations, brackets and word problems

`term`

Second Term

`week`

9

`lesson_objectives`

• carry out order of operation in algebra
• insert/remove bracket and simplify expression
• solve word problems involving algebraic expression

Objective 1 — carry out order of operation in algebra

`objective_text`

carry out order of operation in algebra

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 2 — insert/remove bracket and simplify expression

`objective_text`

insert/remove bracket and simplify expression

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 3 — solve word problems involving algebraic expression

`objective_text`

> solve word problems involving algebraic expression

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Output — the pack it produced (draw 3)

Objective 1

Simplifying Expressions with the Order of Operations

> EXPLANATION
> When you simplify an algebraic expression you must follow the order of operations (often remembered as PEMDAS):
>
> 1. **P – Parentheses (brackets)**: Work inside the innermost brackets first.
> 2. **E – Exponents**: Evaluate any powers or square-roots.
> 3. **M/D – Multiplication and Division**: Carry these out from left to right.
> 4. **A/S – Addition and Subtraction**: Finally, do these from left to right.
>
> In algebra the same rules apply to variables as they do to numbers. Treat a variable like a number when you multiply, divide, add or subtract it. For example, in the term `3 × x` you first multiply the coefficient `3` by the variable `x`.
>
> Remember to keep the steps clear: simplify inside brackets, then handle any exponents, then work through any multiplication or division, and finish with addition and subtraction.

> WORKED_EXAMPLE
> **Worked example** – Simplify the expression:
>
> \[4 + 3 \times (2x - 5) - 2^2 \]
>
> Step 1 – Brackets: simplify inside the parentheses.
> \[2x - 5 \] is already as simple as it gets, so we keep it.
>
> Step 2 – Exponents: evaluate \(2^2\).
> \[2^2 = 4 \]
>
> Now the expression looks like:
> \[4 + 3 \times (2x - 5) - 4 \]
>
> Step 3 – Multiplication: multiply the 3 by the bracketed term.
> \[3 \times (2x - 5) = 6x - 15 \]
>
> Replace the product:
> \[4 + (6x - 15) - 4 \]
>
> Step 4 – Addition and Subtraction (left to right):
> \[4 - 4 = 0 \]
> So we have:
> \[0 + (6x - 15) = 6x - 15 \]
>
> **Result:** \(6x - 15\).

1. Simplify: \[6 + 2 \times (3 + 4) \]
answer: 20

2. Simplify: \[5x - 2(3x - 4) + 7 \]
answer: -x + 15

3. Simplify: \[(8 - 2)^3 \div 4 \]
answer: 54

4. Simplify: \[3a + 4b - 2(a - b) + 5 \]
answer: a + 6b + 5

Objective 2

Removing and Adding Brackets to Simplify Expressions

> EXPLANATION
> In algebra, brackets tell us that the terms inside must be dealt with together.
>
> **Removing brackets** – Use the distributive law. If a number or variable is in front of a bracket, multiply it by each term inside the bracket. Remember to change the sign of each term when the bracket is preceded by a minus sign.
>
> **Inserting brackets** – Look for a common factor that appears in every term. Factor that common factor out and write the remaining terms inside a new bracket. This is the reverse of the distributive law.
>
> Both operations help to rewrite an expression in a simpler or more useful form, and they are essential before you can combine like terms or solve problems.

> WORKED_EXAMPLE
> ### Example 1 – Removing brackets
> Simplify $4(2x - 3) + 5x$.
>
> 1. Distribute the 4: $4 \times 2x = 8x$ and $4 \times (-3) = -12$.
> 2. Replace the original part: $8x - 12 + 5x$.
> 3. Combine like terms: $8x + 5x = 13x$.
> Result: $13x - 12$.
>
> ### Example 2 – Inserting brackets (factoring)
> Rewrite $6y + 9$ as a product of a common factor and a bracket.
>
> 1. Identify the greatest common factor of the terms: both are multiples of 3.
> 2. Factor out 3: $6y = 3 \times 2y$ and $9 = 3 \times 3$.
> 3. Write the expression: $3(2y + 3)$.
> Result: $3(2y + 3)$.

1. Simplify $3(a + 4) - 2a$.
answer: a + 12

2. Simplify $5(2b - 3) + 4b$.
answer: 14b - 15

3. Write $8c + 12$ as a product of a common factor and a bracket.
answer: 4(2c + 3)

4. Simplify $-2(3d + 5) + d$.
answer: -5d - 10

Objective 3

Solving Word Problems Using Algebraic Expressions

> EXPLANATION
> In a word problem you first decide what the unknown quantity is and give it a letter, usually \(x\). Then you translate the words into an algebraic expression, being careful with the order of operations and any brackets.
>
> **Steps**
> 1. **Identify the unknown** and assign a symbol.
> 2. **Write an expression** for each part of the problem, using the symbols and the operations described (addition, subtraction, multiplication, division).
> 3. **Combine the expressions** into an equation that represents the whole situation.
> 4. **Simplify** the equation using the rules of algebra – combine like terms, apply the distributive law, remove brackets, and follow the order of operations (PEMDAS).
> 5. **Solve** for the unknown by isolating the variable.
> 6. **Check** your answer by substituting it back into the original situation.
>
> Remember: brackets indicate that the operations inside them must be done first, and any coefficient in front of a bracket must be multiplied through (the distributive law).

> WORKED_EXAMPLE
> **Worked Example**
>
> *Problem*: A garden is 5 metres longer than twice its width. If the perimeter of the garden is 54 m, find the width.
>
> *Solution*:
> 1. Let the width be \(x\) metres.
> 2. Then the length is \(2x + 5\) metres.
> 3. The perimeter of a rectangle is \(2(\text{length} + \text{width})\). So
>
> \[2\big((2x + 5) + x\big) = 54\]
>
> 4. Simplify inside the brackets: \((2x + 5) + x = 3x + 5\).
> 5. Multiply by 2: \(2(3x + 5) = 6x + 10\).
> 6. Set the equation: \(6x + 10 = 54\).
> 7. Subtract 10 from both sides: \(6x = 44\).
> 8. Divide by 6: \(x = \frac{44}{6} = \frac{22}{3} \approx 7.33\) metres.
>
> *Answer*: The width of the garden is \(\frac{22}{3}\) m (about 7.33 m).

1. A rectangle is 3 m longer than its width. Its area is 70 m². Find the width of the rectangle (in metres).
answer: 7

2. Sam buys 4 notebooks and some pens. The total cost is $23. Each notebook costs $5 and he bought 3 pens. How much does each pen cost (in dollars)?
answer: 1

3. A number is 6 less than three times another number. If the sum of the two numbers is 34, what is the larger number?
answer: 24

4. The sum of three consecutive even numbers is 48. What is the smallest of these numbers?
answer: 14

C1 `The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Simplification of Algebraic Terms

`subtopic`

Simplification of algebraic expressions: order of operations, brackets and word problems

`term`

Second Term

`week`

9

`lesson_objectives`

• carry out order of operation in algebra
• insert/remove bracket and simplify expression
• solve word problems involving algebraic expression

`entry_behaviour`

• Addition and subtraction of algebraic terms
• Multiplication and division of algebraic terms

`instructional_materials`

• Chart showing algebraic expression.

Objective 1 — carry out order of operation in algebra

`objective_text`

carry out order of operation in algebra

`current_step`

1

`instructional_events`

- provide_learning_guidance
- elicit_performance
- provide_feedback

`student_tasks`

- 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$

`assessment_points`

- Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$
- Simplify $7 \times 3a - (3a + 5a) \times 2$

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — insert/remove bracket and simplify expression

`objective_text`

insert/remove bracket and simplify expression

`current_step`

2

`instructional_events`

- provide_learning_guidance
- elicit_performance
- provide_feedback

`student_tasks`

- (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$

`assessment_points`

- Simplify $12 - (7 - 2f) - (3 - f)$
- Simplify $(8a - 5) + (14 - 3a)$

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Objective 3 — solve word problems involving algebraic expression

`objective_text`

solve word problems involving algebraic expression

`current_step`

3

`instructional_events`

• provide_learning_guidance
• elicit_performance
• provide_feedback

`student_tasks`

• 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.

`assessment_points`

• A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Simplifying Algebraic Expressions: Order of Operations

> EXPLANATION
> When simplifying algebraic expressions you must follow the same order of operations used with numbers (often remembered as BODMAS):
>
> 1. **Brackets** – work inside any brackets first.
> 2. **Orders** – handle exponents and roots.
> 3. **Division and Multiplication** – carry these out from left to right.
> 4. **Addition and Subtraction** – finally, perform these from left to right.
>
> In algebra the symbols (letters) behave just like numbers when you apply these rules. Multiply or divide the coefficients and combine like terms only after the operations have been carried out in the correct order.

> WORKED_EXAMPLE
> **Worked example**
>
> Simplify $4n + 3n \times 10$.
>
> 1. Do the multiplication first: $3n \times 10 = 30n$.
> 2. Now add the like terms: $4n + 30n = 34n$.
>
> So $4n + 3n \times 10 = 34n$.

1. Simplify $4n + 3n \times 10$.
answer: 34n

2. Simplify $32 + 8x \div 2$.
answer: 32 + 4x

3. Simplify $3 \times 5x + 4x \div 2$.
answer: 17x

4. Simplify $36y \div 6 + y + 2 \times 4y$.
answer: 15y

5. Simplify $14pq \times r$.
answer: 14pqr

6. Simplify $12x^2y \div 3x$.
answer: 4xy

7. Simplify $7 \times 3a - (3a + 5a) \times 2$.
answer: 5a

Simplifying Expressions by Adding and Removing Brackets

> EXPLANATION
> When you have brackets in an algebraic expression you can either remove them or introduce them to make the expression easier to work with.
>
> **Removing brackets**
> - If a plus sign \(+\) is in front of a bracket, simply drop the brackets and keep the signs of the terms inside unchanged.
> - If a minus sign \(-\) is in front of a bracket, drop the brackets and change the sign of every term inside the bracket.
>
> **Inserting brackets**
> - You may place brackets around a group of terms you want to treat as a single unit, for example when you want to show the order in which operations should be carried out.
>
> **Steps to simplify**
> 1. Remove any brackets by applying the sign in front of them.
> 2. Combine like terms (terms that have the same variable part).
> 3. If needed, rewrite the result with brackets to make the structure clear.

> WORKED_EXAMPLE
> **Example 1**
> Simplify \((12a - 7) + (4 - 5a)\).
>
> 1. The brackets are preceded by a plus sign, so we can drop them: \(12a - 7 + 4 - 5a\).
> 2. Combine the \(a\)-terms: \(12a - 5a = 7a\).
> 3. Combine the constant terms: \(-7 + 4 = -3\).
>
> Result: \(7a - 3\).

> WORKED_EXAMPLE
> **Example 2**
> Simplify \(12 - (7 - 2f) - (3 - f)\).
>
> 1. Distribute the minus sign over each bracket: \(12 - 7 + 2f - 3 + f\).
> 2. Combine the constant terms: \(12 - 7 - 3 = 2\).
> 3. Combine the \(f\)-terms: \(2f + f = 3f\).
>
> Result: \(2 + 3f\).

1. (a) $(12a - 7) + (4 - 5a)$
answer: $7a - 3$

2. (b) $(8x - 2y) - (9x - 4y)$
answer: $-x + 2y$

3. Simplify $12 - (7 - 2f) - (3 - f)$
answer: $2 + 3f$

4. Simplify $(8a - 5) + (14 - 3a)$
answer: $5a + 9$

Objective 3

Solving Word Problems Using Algebraic Expressions

> EXPLANATION
> To solve a word problem with algebraic expressions, first decide what each unknown stands for. Write an expression for each piece of information given, then combine the expressions using the appropriate operations (addition, subtraction, multiplication, division). Finally, simplify the expression to find the value you need.

> WORKED_EXAMPLE
> Example:
>
> A pen costs \(h\) naira and a crayon costs \(20\) naira less than a pen.
>
> 1. Let the cost of the crayon be \(c\).
> 2. From the wording, \(c = h - 20\).
> 3. The total cost of a pen and a crayon is \(h + c = h + (h - 20) = 2h - 20\).
>
> So the crayon costs \(h - 20\) naira and together they cost \(2h - 20\) naira.

1. A pen costs \(h\) naira and a crayon costs ₦20 less than a pen. What is the cost of the crayon?
answer: h - 20

2. Using the same information, what is the total cost of a pen and a crayon together?
answer: 2h - 20

3. Envelopes cost \(d\) naira per pack. Four notepads cost ₦20 more than a pack of envelopes. What is the total cost of one pack of envelopes and one notepad?
answer: (5d/4) + 5

4. A pen and a pencil together cost \(x\) naira. If the pen costs \(y\) naira, what is the cost of three pencils?
answer: 3x - 3y

Result — per rubric dimension

Scores are 1–5, each the mean of 2 judge pass(es) on the same packet. The first three columns are the packs shown above (draw 3); the last is the lesson's mean over its 3 draws per condition — the study's number.

Rubric	C0	C1	C1 – C0, shown packs	C1 – C0, mean of 3 draws
Objective and curriculum alignment	3.00	5.00	**+2.00**	**+2.00**
Taught-lesson alignment	2.00	5.00	**+3.00**	**+2.67**
Scope and conceptual correctness	3.00	4.00	**+1.00**	**+2.00**
Student-facing clarity and usefulness	3.00	5.00	**+2.00**	**+1.67**